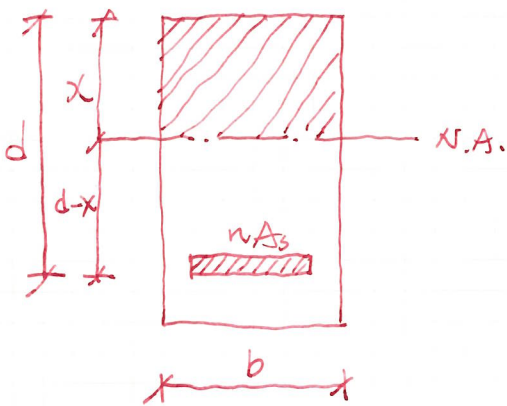


RC-2023-1 補充

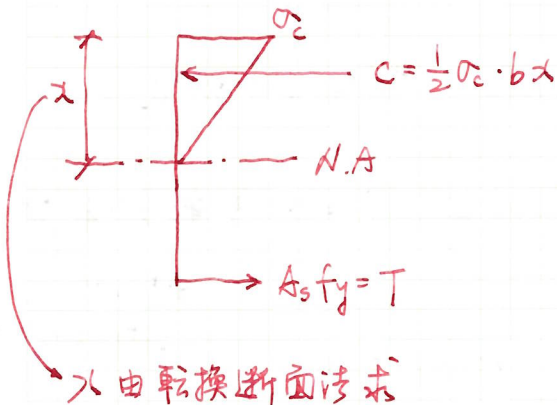
中性軸： $\left\{ \begin{array}{l} \text{力平衡 } C=T \\ \text{材力在平截面假設，斷面上下面積的一次矩相等，只有在線性彈性下才成立，} \\ \text{轉換斷面法} \end{array} \right.$

降伏時：受壓區呈線性彈性，符合材力，可以利用面積一次矩求解中性軸，利用轉換斷面法

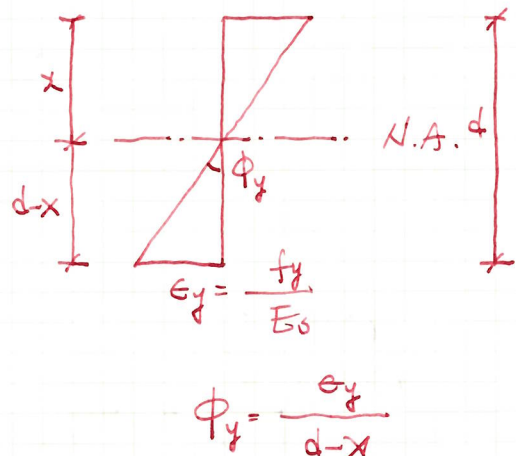


$$(bx) \cdot \frac{x}{2} = nA_s (d-x)$$

降伏應力圖：

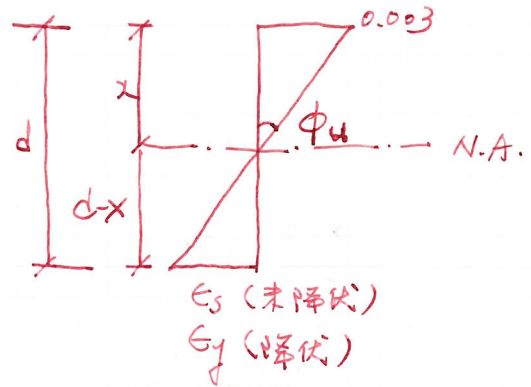


降伏應變圖

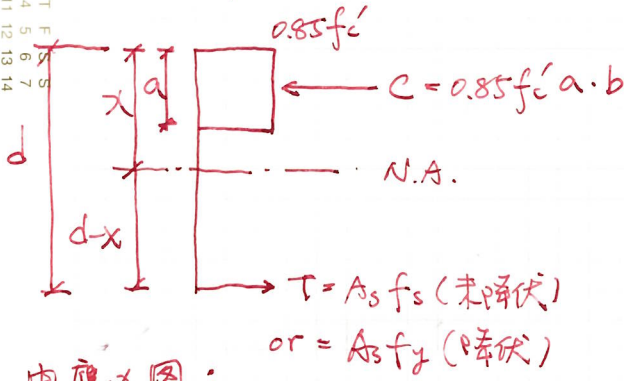


3 MARCH
1 2 3 4 5 6 7
8 9 10 11 12 13 14
15 16 17 18 19 20 21
22 23 24 25 26 27 28
29 30 31

極限應變圖.



極限應力圖.



由應力圖:

$$a = \beta_1 x$$

求中性軸: $C = T$ 力平衡

$$0.85f'_c a \cdot b = A_s f_y \quad (\text{先假設降伏})$$

$$a = \frac{A_s f_y}{0.85f'_c \cdot b}$$

$$x = \frac{a}{\beta_1}$$

由應變圖:

$$\phi_u = \frac{0.003}{x}$$

$$\text{延展比 } \mu = \frac{\phi_u}{\phi_y}$$

討論:

$$f'_c \uparrow, a \downarrow, x \downarrow, \phi_u \uparrow, \mu \uparrow$$

先鋼筋降伏, 再混凝土達應變極限, 是有先後次序